What Can Requirements Engineering Do for Emerging System of Systems? Case of Smart Local Energy

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ABSTRACT

Abstract: As software solutions permeate the whole spectrum of human activities, many software systems previously developed to address specific issues suddenly find themselves becoming parts of interlinked and inter-dependent complex systems. This paper discusses a case study of such emergent convergence of software-rich socio-technical systems into an interconnected and integrated energy system-of-systems and outlines some areas where the Requirements Engineering discipline can help guide such a transition.

Lay Abstract: Software is becoming integrated in all areas of human activities (e.g., bus timetabling and home heating). Solutions that support such distinct activities are now becoming interlinked into complex systems which impact each other (e.g., both electric bus charging and home heating solutions need to balance with electricity grid). This paper discusses a case study of such emerging interdependence for energy systems and outlines where the Requirements Engineering discipline can facilitate better integration.

CCS CONCEPTS

• **Software and its engineering** → Design patterns; **Ultra-large-scale systems**; • **Human-centered computing** → *Empirical studies in collaborative and social computing*; • **Applied computing** → *Cross-organizational business processes.*

KEYWORDS

System-of-systems, Requirements engineering, case study, smart energy system, virtual system-of-systems, systems transition, energy transition

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1 INTRODUCTION

As software solutions permeate the whole spectrum of human activities, many software systems previously developed to address specific issues suddenly find themselves becoming parts of interlinked and inter-dependent complex systems. For instance, a bus timetable monitoring app (developed to support transportation requirements) and a thermostat controlling software (designed to keep comfortable temperature levels at homes) may at a later time be integrated into a smart energy system. Such a smart energy system would ensure sufficient electricity supply at all times by scheduling the timetable for charging the electric buses away from the peak times when householders need to heat up their homes using electric heaters.

This trend of software systems converging into larger *systems of systems* (SoS) is now ubiquitous (e.g., personal exercise apps integrating with doctors' health records and insurance service provision; flight bookings with hotel reservations and health testing). Yet, to date, the software and requirements engineers ***have not assumed a definitive role*** *in guiding the transition of software-intensive socio-technical systems into interconnected and integrated socio-technical system-of-systems.* Neither do they themselves or the relevant converging businesses appreciate the knowledge, methodological grounding and tool support that the discipline of requirements engineering (RE) can contribute to supporting productive SoS emergence. So, the main question addressed in this paper is: *How can RE support the emerging convergence into SoS?*

This paper provides a case study that *observes one such newly forming SoS* in the context of *smart local energy systems.*

The study demonstrates that this convergence into a SoS:

- has *no clear owner or governance* in place and lacks guidance;
- is an *innovation and co-creation process*, whereby quickly developing new technologies enable new types of energy services and business models. Thus, the requirements of the digital services cannot be elicited ahead of the emergence of various SoS properties and characteristics.
- results in a *new type of socio-technical systems development* process where all stakeholders - from businesses to local and national authorities and individual citizens - are engaged with the SoS construction effort.

As a result, it is not really possible to undertake a "usual" requirements engineering process, where the system owners and users could drive the requirements specification. Instead, this process *re-defines the role of a requirements engineer from that of "elicitation" to that of "orchestrating requirements co-creation".*

Despite this radical change of the requirements engineer's role, a wide variety of RE methods and techniques can be drawn upon to support various aspects of this SoS construction activity. Yet, we observe that for such an (un-planned) emergent SoS construction to succeed, an even more important part of the requirements

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engineer's job is in bringing the key stakeholders together and supporting their collaboration in:

- building a coherent, unified vision of and goal for this virtual SoS, while also
- fostering the multilateral agreements on specific services, policies and practices between stakeholder sub-groups.

The key contributions of this work are in:

(1) Illustrating the socio-technical systems convergence into a system of systems through an example of the Smart Local Energy transition (section 4);
(2) Sharing the methodology for analyzing such a transition (section 3);
(3) Reporting on challenges observed within the present case study that a requirements engineer should address and discussing some examples of RE approaches and practices that can be applied to help address these challenges (section 5).

The rest of this paper is structured as follows: section 2 sets the background to this work by discussing the notion of a System of Systems and how it relates to smart local energy (SLE) as well as the related work on Requirements Engineering for SoS. Section 3 outlines the case study context, which is followed by the details of the research design, participant recruitment and data analysis methods. Section 4 presents the study findings, presenting examples of sub-system specific analysis and the SoS level integrated view for smart local energy system. Section 5 discusses the challenges that SLE faces in its' emergent convergence towards a SoS and notes how the contemporary RE research can contribute towards addressing these challenges. Finally, section 6 concludes the paper.

2 BACKGROUND AND RELATED WORK

2.1 Smart Local Energy as a System of Systems

Smart local energy (SLE) solutions aim to employ software to manage and balance local (i.e., geographically bound) [28] energy supply, storage and use across all energy vectors (e.g., power, heating and transport), bringing efficiencies to energy supply and demand, and social, environmental, and economic benefits to the given locality. A rich picture of an example SLE system is presented in Figure 1, additionally noting that:

- Given the zero-carbon economy targets for 2050, energy generation must transition to green and renewable-based sources.
- These sources are *distributed across localities*, e.g., tidal energy can only be harvested on the shore side, etc.
- Finally, renewable generation technologies are becoming widely adopted both by households and businesses who wish to generate clean energy for their own use. Thus, energy *distribution and decentralization* are a key characteristic of SLE systems.

The *"smart"* aspect of energy systems implies *digitally supported coordination of decision-making* to optimize resource use and waste reduction (both in generation and consumption), fault tolerance and recovery from failures, and support for human decision-making for efficiency and comfort. On the other hand, a smart energy system will not fulfill its potential without *smart users*, thus household and business users also need to acquire skills in the functioning and use of digital energy systems [11, 15, 17, 70].

2.2 SLE as a System-of Systems

Smart local energy is itself *a system of systems* (as shown in Figure 1[1]). As per the SoS definition [49], a set of individual sub-systems (such as local/national government, energy supply, building and retrofit provision, transportation and mobility services provision, local citizens and communities, the community energy groups and the ICT infrastructure for digital energy services) are to be to some degree integrated (e.g., through a digital and physical infrastructure). These individual sub-systems have their own boundaries and behaviors, but, in becoming a part of SLE, they must collaborate towards a common goal (such as the goal of optimal use of local energy, carbon emissions reduction across a locality, etc.). In collaborating towards this goal, the sub-systems exchange data and exercise some control over the SoS's operation and data exchange.

SoS are often defined by the set of traits that they exhibit [49, 51, 66]. Based on these traits (e.g., the degree of operational and managerial independence of sub-systems and their evolutionary development, all of which defines the degree of the system's centralized control and autonomy), a SoS is generally categorized into 1 of 4 types [49, 66]:

(1) A *Virtual SoS*, has no centralized authority directly overseeing its sub-systems interactions. The interaction and respective emergent behaviors occur through indirect and invisible mechanisms (e.g, mechanisms such as interest rates and supply and demand regulate the national economy SoS, the sub-parts of the SoS do not formally agree to collaborate);

[1]Note that a number of relevant sub-systems, such as manufacturing and finance, are omitted from this rich picture for simplicity. These sectors, while very important to SLE operation, do not require any dramatically new solutions for SLE, as per our current case study analysis. However, in discussing the presented sub-systems, the topics related to those omitted ones are also addressed, as relevant.

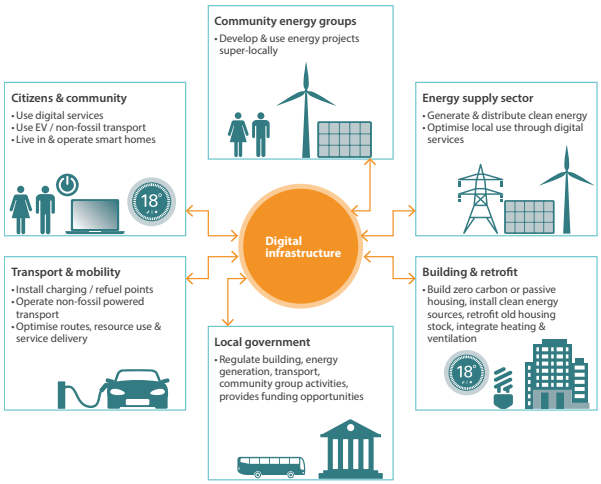

Figure 1: A (simplified) rich picture of a smart local energy system

(2) In a *Collaborative SoS* the sub-systems interact on a voluntary basis and collectively decide on how to provide a particular service (e.g., the internet is composed of collaborating national networks);
(3) In an *Acknowledged SoS* all sub-systems acknowledge their common objectives and work through a designated management team to coordinate resource use and plans for achieving the agreed goals (e.g., as is done in a military command and control SoS);
(4) In a *Directed SoS* the constituent sub-systems are well integrated, are managed by a centralized team/authority and operate as the subordinates of this central authority (e.g., as the centralized electricity system has been until recently).

2.3 Related Work on SoS

Previous research that studied requirements for SoS has been done both in RE and Management Science.

2.3.1 ***Related Work in Requirements Engineering.*** Dahmann [26] enumerates the main "pain points" of RE for System of Systems work as: authorities of the SoS, leadership needs, ways of integrating constituent sub-systems, engineering requirements for SoS capabilities and requirements, addressing autonomy and emergence as well as testing and validating, and finally eliciting key principles behind the SoS paradigm. In addition, Ncube et al. note a need for tools and metrics development, security and architectural requirements, multi-level modeling and virtualization as areas where RE research and development must innovate [52] to handle SoS needs.

Lewis et al. [48] suggest an abstract method which would use "a combination of top-down and bottom-up approaches" (i.e., engineering requirements for the sub-systems as well as for the SoS as a whole) through a set of activities. These comprise analyzing (i) SoS context (i.e., centralized vs. distributed operation; entities and their influence and interactions), (ii) SoS and individual system goals (e.g., using goal modeling to find common and conflicting goals), (iii) SoS interactions (e.g., use cases and scenario modeling techniques to observe process, data, and resource-centeric interactions), (iv) individual system capabilities and constraints, and (v) the gap between individual systems and the SoS, comparing the top-down system requirements (i.e., goals, context and interaction) and bottom up sub-system capabilities and constraints.

Walker et al. [67, 68] demonstrate an implementation of a method similar to Lewis et al. [48] in the context of the defense sector. Here, after establishing the boundary of the prospective SoS (e.g., using a rich picture), two parallel streams of activities (top-down and bottom-up) are suggested. The results of these activities would then be compared to identify which sub-systems will provide which functionalities for the SoS; highlight the functionality duplicated through several sub-systems either by design (e.g., for failure tolerance) or by accident; and identify any functionality expected for the SoS but not currently addressed by any of the sub-systems. The requirements are then formulated with respect to the required functionality provision and constraints. However, this method presumes the availability of rather detailed documentation (e.g., aircraft operational specifications, etc.), which may not be readily available for most non-defense SoS domains.

Holt et al. [40] present a model-based approach which represents requirements at both the sub-system and the SoS levels, along with the specific SoS contexts and constraints. Yet, this solution is focused on the pre-agreed functionality and pre-set contexts that are validated and enforced for the SoS. It does not accommodate for the emergence and complexity of a real, operational SoS and also assumes that correct and rather complete requirements have already been somehow elicited.

Naqvi et al. [51] propose a natural language processing-based approach for the dynamic identification of prospective interactions within a SoS. The interactions of interest are to be represented through composition specifications using an NLP-based composition language. Once specified, these compositions would identify similar interactions in any new or changed requirements documents with no further analysis effort. Here too, the authors assume that the kinds of emergent interactions are known and requirements documents of some kind available. However, when and where such interactions will emerge does not need to be pre-defined.

Thus, we note that the solutions proposed for RE for SoS, so far, either expect detailed knowledge and documentation of the sub-systems and SoS as a whole [40, 51, 67, 68] (i.e., looking mainly at directed SoS), or rely on high-level guideline recommendation [48]. Case studies with methods supporting analysis of the newly emerging (most particularly) virtual and collaborative SoS sub-types are presently scarce.

2.3.2 ***Related Work in Management Science.*** To address the challenge of SoS risk management, Tundera et al. [63] suggest 3 basic principles: (i) implementing the different levels of responsibility for management and (social) impact mitigation; (ii) integrating the SoS behaviors management with national politics and economic development policies; and (iii) promoting relevant technology, modernization, and legalization.

Zhu et al. [75] propose a *multi-level* framework for SoS management. It accounts for factors at *base level* (e.g., requirements elicitation by an individual analyst), *activity level* (i.e., activity of requirements elicitation at the company for a project), *process level* (e.g., requirements elicitation and how it is integrated with product design, etc.) and *project level* of the SoS (e.g., managing a whole IT project) by measuring *human* (e.g., skills and risk attitudes), *resource* (e.g., quality of materials, cost, etc.) and *information* (e.g., priority and accessibility) attributes at these different levels. This assumes that all above attributes can be identified and measured across the SoS - a challenge on its own, if considering the SLE SoS (see Fig. 1).

Gunawan et al. [35] propose a SoS management framework to help systems engineering managers to guide SoS projects. This comprises: (i) constructing a brief description of the SoS; (ii) compiling descriptions of external factors that could affect the subsystems or the whole SoS (e.g., environmental protection, local economic development, etc.); (iii) collecting information on the feedback provided by the SoS (e.g., public perception affected by public media could enforce local policies, etc.); (vi) detailing what the governing body of the SoS is; (v) describing the constraints that could affect the governing body; (vi) describing how the governing body directs the SoS. The impact of various factors on the SoS is analyzed qualitatively through causal loop diagrams and quantified through system dynamics models.

Research in Management, similar to that of RE, tends to expect the presence of a manager or management body who is interested in using guidelines or detailed data, metrics and measurements for managing the SoS formation. This is clearly not the case within the virtual and collaborative SoS sub-types, of which SLE is one.

While all of the discussed research contributes to the body of evidence for RE practice for the SoS, none of these provide any useful guidance on *how to undertake RE for a large SoS, such as a smart local energy system, where no clear governing body is available and the sub-systems are only just (if at all) starting to realize that some kind of convergence is underway.* The present paper provides an experience-based report, aiming to facilitate such an activity for forthcoming research and practice.

3 STUDY DESIGN

3.1 Case Study and Research Question

The key question addressed in this paper is: *How can RE support the emerging systems convergence into SoS?*

This question occurred as the author - a requirements engineer - was grappling with the challenge of supporting the *city-wide transition to SLE* as part of a research collaboration with the Bristol City Council (BCC). However, this question was not the focus of the Bristol City SLE transition case study analysis. Instead, the case study was *concerned with the Bristol city's transition to SLE, aiming to understand the drivers, obstacles, and needs of each stakeholder type (e.g., energy sector, transportation sector, etc.) as well as the SLE system as a whole.* This is natural, as only after understanding what the emerging SoS concerns are, one can consider how these could be supported by the RE discipline. Accordingly, the specific questions addressed throughout this study are listed in Fig. 2.

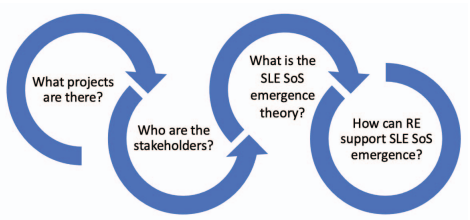

Figure 2: Questions addressed in this study

The research that addresses these questions is detailed below.

3.2 Study Design

As detailed below, for this study we undertook:

(1) A review of grey literature and materials to identify the SLE projects undertaken in the Bristol city;
(2) The identification of per-project stakeholders;
(3) Qualitative data collection (via interviews, focus groups, reflective writing) to elicit the SLE transition theory.
(4) Consideration of RE support for each of the key facgtos for the SLE emergence theory (see section 5).

3.2.1 ***What are Bristol City's SLE Projects?*** A review of grey literature was carried out to identify past and present SLE projects held within the city of Bristol. The review included:

- Web materials (websites, online reports, white papers, etc.) from the prominent local stakeholders within the energy sector (e.g., from Bristol City Council, Bristol Energy Network, etc.). We also integrated the stakeholders identified as relevant to the SLE SoS within Bristol city from the dataset on SLE businesses in the UK [31].
- The databases on SLE funded projects from the UK research agency [4].
- A dataset of 119 energy system demonstrator projects collected by previous research [2].

The review showed a picture of Bristol as a SLE transitioning city.

3.2.2 ***Who Are the SLE Stakeholders in Each Project?*** Along with the project identification process, for each project, the set of relevant stakeholders were identified to build up the picture of participants in the Bristol city's SLE scene and their roles. Next, we needed to see how these stakeholders contributed to the SLE scene.

3.2.3 ***What Are the Per-Stakeholder Drivers, Obstacles and Skills Needs (i.e Emergent Theory) for SLE Transition?*** To address this question on drivers, obstacles and skills needs for transition to SLE, the primary qualitative data collection was carried out through:

- Interviews of 24 stakeholder representatives at 23 interviews (one interview was carried out with two interviewees present together);
- Two focus groups with 10 participants overall;
- The writing of one collaborative reflective summary report with 7 participants.
- An interview study with 30 citizens. We note that while SLE service delivery is provided by organizations, the adoption and use of these services is strongly dependent on the citizens. Thus, the study of how and why citizens would engage with SLE ecosystems was deemed to be necessary.

The data collection took place from Oct 2019 to Oct 2020, during the period when COVID-19 restrictions were imposed upon Bristol city. Consequently, some of the interviews were carried out face-to-face, while others, along with the focus groups and reflective report writing, were carried out via videoconferencing.

Data was collected in accordance with the University of Bristol's ethics committee processes and rules. The participants either signed and returned a consent form or (during the COVID-19 restrictions period) were emailed the study materials and consent forms and asked to verbally confirm their consent at the time of interview recording and to return the form by email. In all cases the conversations were recorded and transcribed for qualitative data analysis.

Participant Recruitment for Stakeholder Interviews:

The process of projects and stakeholders identification (discussed in above sub-sections) was carried out iteratively. We found a number of institutions which had participated in more than one such project, which indicated the presence of a longer-term interest and commitment to SLE within these organizations.

These organizations were then invited for (up to 1 hour long) interviews, (2 hour long) focus groups and (asynchronous) collaborative writing session (as show in Table 1), while:

Table 1: Details of Study Participants

ID	Role within Organization	Field of Work	Data Collection
P1	Software Engineer	ICT	FG
P2	SME energy project developer	Energy	Interview
P3	Community project developer	CE	Interview
P4	Regional project manager	LA	Interview
P5	Technical developer, community energy	CE	Interview
P6	Local authority partnerships manager	LA	Interview
P7	Retrofit project manager	B&R	Interview
P8	NGO Community manager	CE	Interview
P9	Energy company Innovation Manager	Energy	Interview
P10	Chief Marketing Officer: Smart Mobility	T&M	Interview, FG
P11	Sustainable Building Consultant	B&R	Interview
P12	Consultant for business case analysis	B&R	Interview
P13	CE project manager	CE, B&R	Interview
P14	Building company managing director	B&R	Interview
P15	Trades union regional secretary	Community	Interview
P16	Trades union education officer	Community	Interview
P17	Community energy development manager	CE, Funding	Interview
P18	Local authority Investment Manager	LA	Interview
P19	Community worker	Community, CE	Interview
P20	ICT for Smart Mobility Researcher	ICT	Interview
P21	Transport tech projects director	T&M	Interview, FG
P22	Funding associate	Community	Interview
P23	Architect	B&R	Interview
P24	DSO and Future Networks Manager	Energy	Interview
P25	Data and Technology Architecture Manager	ICT	FG, Reflect.
P26	Network & Developer Engineer	ICT	FG, Reflect.
P27	Sustainable Neighbourhoods Manager	Community	FG
P28	Associate Professor: ICT	ICT	FG
P29	Project Manager: Smart Energy	LA	FG, Reflect.
P30	SLES Senior Researcher	ICT	FG, Reflect.
P31	SLES Senior Project Manager	LA	FG
P32	SLES Project Researcher	CE	Reflect.
P33	Senior Lecturer: ICT for SLES	ICT	Reflect.
P34	SLES Senior Researcher	ICT	Reflect.

Abbreviations: ICT: Information and Communication Technology, LA: Local Authority, CE: Community Energy, B&R: Building and Retrofit, T&M: Transport and Mobility, FG: Focus Group, Reflect.: Reflective Writing

- Aiming to recruit participants from within organizations with more than one SLE project engagement;
- Engaging specifically those individuals within the organizations who, if at all possible, had themselves been part of more than one SLE project; this was done in the hope of obtaining more experience-based data from such participants;
- Engaging a mix of roles - from those with hiring and project management responsibilities to those who were directly engaged with the project delivery;
- Engaging the whole organizational spectrum of the SLE subsystem categories (as per Fig. 1).

The ***study questions*** cover topics[2] on:

- Participant and organization background details;
- Activities within the SLE domain;
- Drivers and Obstacles in the transition to SLE with reference to the relevant SLE projects;
- Current and future skills and training needed for transitioning to SLE.

[2]All topics were used for the interviews; the focus groups and collaborative report did not address the first two topics as that info was collected ahead of the meetings.

Participant Recruitment for Focus Groups:
Two online (zoom-facilitated) focus groups were carried out (with 10 participants overall) to expand upon the input received from interviews. The focus group participants were drawn from an active SLE SoS project coordinated by the BCC. As noted in Table 1, two of the focus groups' participants had been interviewed previously, while 8 had not. The questions discussed at the focus groups were same as those asked during the interviews, but were also contextualized for the common SLE project.

The focus groups were also transcribed and then analyzed along with the interview study data.

Participant Recruitment for Collaborative Writing:
During the focus group discussions, the study participants suggested that it would be relevant for their project to collect software development team's reflections on SLE transition experiences. To do this, a shared google document was set up post-focus groups and interested parties were invited to note down their reflections. Four of the focus group participants contributed to this document (as shown in Table 1), as well as invited three new stakeholders with whom they had worked on the project and considered relevant for contributing to the reflective writing exercise. This document too, was used as part of the qualitative data analysis.

Participant Recruitment for Citizens' Sub-System Study:
The interview study with the citizens was focused on topics of smart energy services use at home . Specifically, questions set for the householder interviews were focused one what the householders wished a smart energy system to do; what would foster better adoption of such systems; and what skills and training would the householders need for such adoption [19].

The study was piloted [20] and then carried out with 30 individuals from 28 households (with two interviews carried out with couples, a total of 11 male, 19 female).

Interviewees were part drawn from households that had taken part in SLE projects with the BCC (16 in total) and households with no direct relationship to BCC (12 in total). In the participant recruitment, an active effort was made to obtain a representative sample, balancing for both demographic and owned/occupied property characteristics of the households. Detailed analysis of the citizens' interview study is presented in [17, 19] and is not repeated here.

3.3 Data Analysis

The study participants were grouped per sub-domain of SLE (i.e., transport, energy supply, ICT, etc.) based on the focus of their organizations, as shown in Fig. 1). This forming a separate analysis group for each sub-domain was necessary, as each such sub-domain is driven by its own objectives (e.g., deliver goods and people to their destinations on time, supply electricity, etc.), undertakes its own key activities (e.g., transport goods and people), and is constrained by its own rules and regulations.

Thereafter a constructivist strand of the Grounded Theory analysis method [16] was used. This analysis method was selected in keeping with the guidelines suggested by [61], as the initial (broad) research question was set at the beginning of the project and the researchers already had significant previous knowledge of SLE literature which they could not ignore. Thus, we allowed for the unconstrained categories to emerge from the transcribed

texts through line-by-line text analysis in the initial coding stage (e.g., systems integration skills, lack of certainty on subsidies, etc.). The codes were then integrated into a set of main categories during the *focused coding* activity for each of the sub-domains. During the *theoretical coding* the causal relationships between categories were studied and causal loop diagrams (CLD, see Fig. 3) were built for each sub-domain of SLE (all designs are detailed in [18]). To build an understanding of the wider SLE system, the per-domain CLDs were further cross-compared along with their respective qualitative coding categories from which an integrated SLE SoS model for the Bristol city emerged (see Fig. 4 and section 4) along with the theory of SLE SoS integration (see section 5). The discussion on how the SLE SoS transition theory needs can be supported by the RE discipline is presented in section 5.

While a detailed description of the coding is not presented in this paper, a sample extract is presented in Table 2.

3.4 Threats to Validity and Study Limitations

Given that this is a qualitative study based on data obtained through interviews and focus groups, we do not claim that the findings are generalizable beyond the scope of this case study. Given that the results are based on the studied context and collected data, this is an expected limitation.

While further case studies will be designed to add to the evidence that could validate the results obtained from this case study (as indeed is our ongoing work), findings from such additional studies will not alter the validity of the study for this given context.

We have made our best effort to engage with a representative sample of participants for both the interview and focus group studies, however, we note that these are only representative of the community living in the Bristol city and the context of the SLES projects that have been taking place in the city of Bristol, UK.

In addition, the pool of participants was limited to those organizations which were invited to take part in this study. This may contain a certain selection bias. This concern, however, is mitigated by the fact that the invited organizations are also the very same organizations that would likely engage with broader SLES adoption.

To further the validity of our findings, data, investigator, method and theory components are triangulated [32]:

- for data triangulation [32] we reached out across both *space* and *time*. With respect to space triangulation, the interview participants were drawn from across the whole city, reaching out to stakeholders wherever the SLES projects were identified. Data from stakeholders covers both pre-COVID-19 and COVID-19 periods, collected over a one year period.
- for investigator triangulation, two researchers worked on the thematic coding and analysis, continuously double-checking and verifying each other's work as well as discussing and resolving disagreements.
- for data collection method triangulation, we used interviews, focus groups and a reflective writing method, thus a variety of data inputs inform the analysis.

4 FINDINGS ON SLE TRANSITION THEORIES

Each of the sectors discussed in Fig. 1 were analyzed independently for drivers and obstacles in transitioning to SLE [18] from which

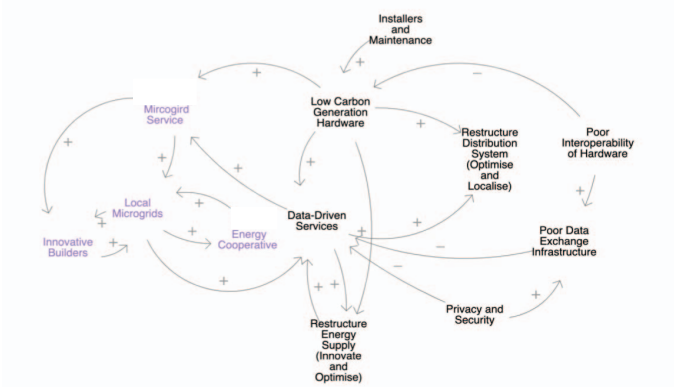

Figure 3: Causal model of Bristol City's Energy Subsystem (factors specific to Bristol City are presented in purple).

sector-specific causal-loop diagrams-based theories of transition were constructed (see section 4.1). These CLDs were cross compared and integrated, from which a holistic picture of SLE transition for the Bristol City emerged, along with its theory (section 4.2).

4.1 Sector-Specific Theories of Transition

To provide an illustration of the sector-specific analysis, we discuss the examples of the Energy/Power Systems sector. The analysis of the other sectors is presented in [18].

4.1.1 Energy Sector. The factors affecting the Energy sector in the Bristol City are illustrated in Figure 3 below. The business representatives interviewed from the energy sector in Bristol City have discussed two key focal factors of convergence to SLE, both driven by local and UK-wide carbon reduction priorities and both leading to the emergence of these new business models: (a) digitization of energy services and (b) investment into new basic assets.

Figure 3 shows that as energy businesses are obliged to diversify from fossil-based generation sources to renewable alternatives (i.e. invest into low-carbon generation hardware), they face the immediate need of engaging with data-driven service delivery. This is necessitated by the:

- Intermittency of generation of these (currently prevalent) non-fossil fuel based sources,
- Short supply and high price of the current energy storage facilities (e.g. electric batteries), and
- Optimization opportunities for the reduction of upfront investment into changing infrastructure (e.g. shifting household consumption away from peak times to avoid investment into the reinforcement of electricity distribution hardware).

However, given the current lack of dedicated infrastructure for collection, storage, exchange, sharing and monetizing of energy sector data, each company wishing to deliver data-driven services needs to set up this type of infrastructure independently. There are many challenges in the setup as it requires, for instance:

- Installation of various data collection and storage hardware;
- Development of software solutions that enable utilization and monetizing of data;
- Provision of interoperability between the various hardware components as well as their software interfaces and services;

Table 2: Coding Extract Examples from Energy Supply Domain Analysis

Initial: Changing Regulation: "regulatory changes that are happening around faster switching which is quite a huge operational overhaul for an energy company. Because it requires a lot of dataflow, optimisation basically..." (P9)	**Focused**: Barriers to Transition/ Changing Regulation	**Theoretical**: Energy supply businesses are obliged to diversify from fossil-based generation sources to renewable alternatives due to Net Zero governmental targets. Thus, the energy companies have to engage with digital energy services (as clear energy sources are intermittent). Yet, as the regulation on energy data is still weak, complicated due to the GDPR, and tends to change, investment into such sources is also risky.	**Theoretical**: Privacy and Security related regulations are perceived as impeding development of digital energy services.	**Theory**: These processes relate to the governmental influence on the SLE transition. This takes place in all sub-systems and can have both positive (e.g.„ the net Zero Targets) and negative (e.g., unclear and changing regulations) impacts.

- Assured checks and balances for regulatory and legal obligations in data use, e.g., for privacy and security concerns.

The above mentioned challenges of substantial investment into assets and tackling data-driven service delivery issues are leading to a deep re-structuring of the Bristol City's energy supply sector (e.g., resulting in selling off Bristol City Energy as a loss making business by the BCC, as well as the sale of the Western Power Distribution company by its American ownership).

On the other hand, the Bristol City has an examples of business emerging whereby the **deployment and management of renewable energy and energy efficiency assets drive the business model without a core focus on the additional digitization services**. For instance, a local microgrid services provider considers digital service provision to be a future possibility. Instead, it is presently focused on the integration of renewables-based generation and consumption technologies (e.g. community-level solar arrays and batteries, heat pumps or district heating technology) into the fabric of the local communal landscape at the construction time. The business then undertakes the management of the energy generation assets, ensuring that the local community benefits from these assets as much as possible while the excess or shortage in supply/demand is purchased from a partner supplier (in this case, Bristol Energy Cooperative).

4.2 Integrated Theory of SLE SoS

The integrated CLD model for accelerated transition to digital services for clean energy for the Bristol City is shown in Figure 4. The model also demonstrates the SLE SoS Transition theory which emerged from our Grounded Theory analysis, suggesting that:

In order to accelerate the transition to clean local energy systems, the wider variety of local sub-systems (including transport and mobility, energy supply, building and retrofit, ICT services, local government, community energy and the citizens) must integrate into a System of Systems. The integration would take place through four key areas and these, consequently, need to be harmonized for a productive SLE SoS. These areas are:

(1) ***Interfaces with Energy Distribution and Transition Networks*** that refer to the hardware-level interlinking of generation and consumption equipment with electricity and gas/heat networks (parts marked as orange in Fig. 4);
(2) ***ICT interfaces*** that integrate data collection, exchange, decision support and control over the various generation, consumption, and storage devices located within the component sub-systems (parts marked as black in Fig. 4);
(3) ***Policy and Regulation*** that constrains and stimulates various activities within these sub-systems (parts marked as green in Fig. 4);
(4) ***People/Education and Training provision***, which fosters engaging with users and other stakeholders within and across the SLES sub-systems (parts marked as purple in Fig. 4).

Looking at the key categories which emerge from the GT analysis (represented as the nodes in Fig. 4), we note that the key defining features of SLE SoS projects are in:

Complexity due to the need to integrate multiple stakeholders and their hardware and software interfaces. This emanates from the fact that SLE SoS projects face the need to integrate across multiple energy vectors (e.g., solar, wind, bio-fuels) and SLES sub-domains (e.g., transport and mobility, household consumption, community generation, digital energy services). Each of these vectors and domains (by necessity) is supported with custom hardware and software solutions. Orchestrating a consistent solution across these heterogeneous sets of hardware and software solutions is a complex challenge by itself. The difficulty is further aggravated by the varied provision of networking and connectivity infrastructure as well as by the fast evolution of renewable-based technologies and their supporting software services.

Networking and Connectivity is essential for data exchange for the monitoring and control that smart energy solutions are expected to exercise within the SLES. Yet, availability of this infrastructure is uneven across the cities. In localities with high income and dense economic activity, the networking provision is assured while in more deprived areas it is scarce.

Given the unprecedented amount of research and development dedicated to zero-carbon SLES technologies, it is not surprising that *technological progress* leads to a fast cycle of hardware and software obsolescence. This fast technological change, in turn, *disrupts convergence to standards* and emergence of stable SLES infrastructures.

It is also noted that *example technical solutions* have a substantial role to play in demonstrating the positive impact and role that ICT-based SLES solutions have. This, in turn, allows other organizations/projects to replicate the aforementioned solution, thus

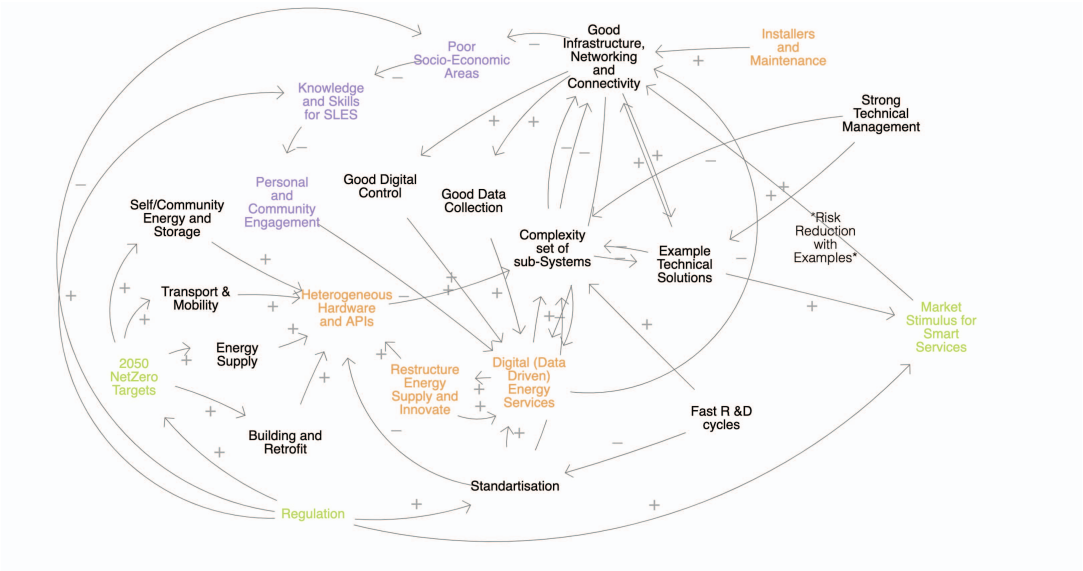

Figure 4: Integrated Causal model of the Bristol City's SLES System.

reducing the complexity that they have to cope with. The growing number of examples also improves the state of connectivity provision, where the examples demonstrate viable business use cases and reduced environmental impacts. Furthermore, such viable use cases also motivate policy makers to provide a market stimulus to foster the replication of such solutions.

In addition, *strong technical management* of SLES projects (i.e., where the project manager is well familiar with the technical issues and engineering technologies of SLES) helps to reduce the complexity of the project and deliver a successful sample solution.

Where the intended user communities are knowledgeable about and engaged with SLES (e.g., with community energy groups or co-designing the prospective digital SLES services, etc.), the likelihood of successful adoption and use of SLES solutions increases (as demonstrated by the EnergyREV project in Bristol City, for example) and with increased user demand the respective services and infrastructure supplies grow and improve as well.

Finally, we observe that policy and regulation have a strong role to play in fostering technological progress through:

(1) enforcing standards (e.g., on hardware and software APIs);
(2) mandating new technology adoption (e.g., the BCC's mandate for 20% renewable energy per new building fostered solar PV and heat pump adoption);
(3) providing funding for user education and engagement as well as stimulating new market models and businesses.

5 DISCUSSION ON RE CONTRIBUTIONS

Drawing on the integrated theory for SLE SoS Energy Transition for the Bristol City's case study, we now discuss and highlight the particular issues within the 4 noted integration aspects in the emerging (mostly virtual) SoS which are likely to be of relevance to requirements engineers. We also point out where and how some of the existing RE research can be relevant.

5.1 Physical Infrastructure Integration

In order to operate an integrated system, the sub-systems within the SLE SoS must support **physical interconnections** for energy (e.g., electricity, heat, gas) exchange.

In most cases such infrastructure is developed along with the renewable technology installations (e.g., solar PVs are integrated with the electricity network at installation time, as are EV charge points and heating and ventilation equipment in buildings, etc.).

Given that a wide set of physical devices is being integrated, a requirements analyst would consider:

- The current and(prospective) future *ownership of these devices* (for instance, in the present energy system an electric vehicle can be owned by a taxi fleet operator or an individual household). Difference in ownership structures will drive different business models and system configurations.
- The *technical impact and likely need of at-scale integration* of each type of device (e.g., integrating a single EV has no impact on an energy system, but integrating a taxi fleet may overstep the capacity of the local energy distribution station and require an upgrade to the distribution station itself).
- *Technological constraints* imposed by each technology (e.g., the material structure of the current gas distribution network corrodes when used for new gaseous fuels such as hydrogen).

Many of the above discussed dependencies are explicit within specific sub-systems, though their relevance to the wider SoS is not always acknowledged. Addressing these issues requires planning, coordination, testing and communication across the SoS sub-systems.

Related work on RE that could facilitate the SLE transition process here include eliciting requirements for infrastructure restructuring [13, 23] and support of data-driven service delivery [33, 56, 74], as well as providing clear requirements for hardware integration and network security [45].

5.2 Sub-Systems Data and Communications

Within the SLE SoS *exchanging data and information* between sub-systems is critical for decision support, control, and optimization.

When digital infrastructure and data are involved, the requirements analyst is suggested to consider that:

- Collection and/or sharing of energy data may *not be a key requirement for any of the sub-systems* themselves. Yet, this is *of paramount importance for the SoS as a whole*.
- Thus, requirements for *data collection and control infrastructure* (such as installation of telecommunication networks, development of software platforms and APIs for data exchange and support for external control functionality) should be considered at the SoS level.
- In addition the *policy and regulatory constraints around data and control* must be defined, as well as the processes through which these are going to be monitored and enforced.

Much of this can be directly supported by **related work on RE**. For instance, drawing on such research as specification and compliance checks with data interchange standards [8] and requirements [53], and various common processes and legal compliance requirements, such as stipulating requirements for the GDPR and storage and sharing constraints as well as data protection [39]. Tools and techniques for extracting legal requirements from regulatory and systems description texts [36] along with the common templates developed for them [59] are also invaluable.

5.3 Policy and Governance of SLE SoS

For RE purposes, we note that all SLE SoS sub-systems:

- interact through energy supply and demand mechanisms and thus factors affecting energy price would be relevant to consider. Oftentimes in a virtual SoS (as is the case in the present-day SLE SoS) the individual sub-systems are not even aware of each other's relevance to the SoS. Indeed, in the present case study, the study participants demonstrate a *severe lack of a holistic view of the SLE SoS*. Building such a common view of the SLE SoS is a key step in engineering a successful SLE SoS. Thus, the requirements that should be considered also include: building such a common view, setting up an effective *SoS governance*, supporting collaboration and *conflict resolution between the SLE SoS parties*.
- want to utilize digital energy services and therefore the delivery of such services and their operation are relevant;
- are also driven by environmental impact reduction, thus regulations and policies on environmental impact as well as the personal beliefs and attitudes of intended users and stakeholders are relevant.

The lack of a holistic view is a common phenomenon in RE and can be supported though **related work** which is either helping relevant stakeholders gain a better understanding of the problem, or is allowing each relevant stakeholder to focus on addressing their part of the problems and (if relevant) combining these partial views to form the full system at a later time [30, 60]. While SoS researchers [26, 52] point out that due to the scale and complexity of a SoS (as confirmed by the SLE example) it is not possible to obtain and maintain a complete understanding of the whole SoS, our notion of a "holistic view" neither aims for nor claims "complete knowledge", instead we suggest these needs:

Requirements for a common SLE SoS goal, agreed upon by (most) stakeholders, would need to be identified. Setting a common goal in a virtual SoS is not an easy endeavor: as previously noted, each sub-system has its own goals to pursue and little interest in the goals of the others. Thus, requirements for wide stakeholder engagement and consensus building process would be needed.

Turning to the present case study, we note that the BCC has already embarked upon this process. In 2019 the BCC initiated the Bristol City's One City Plan [1] which defines how the stakeholders within and outside of the city will work together to create a 'Fair, healthy and sustainable city'. This plan was developed through extensive *engagement and consultation with citizens and wider stakeholders* and provides a set of agreed-upon goals through a collective vision for organizations and individuals across the city, not just for the city council. It is through this engagement that the city has set its net zero targets and all of the Bristol City's sub-systems are prompted to address these targets.

Furthermore, the BCC itself is a critical stakeholder (as it owns about 40% of the land in the city), it *sets a strong trend* by stipulating environmentally-focused constraints on its property developers (e.g., 20% of energy to be renewables-supplied for each new building, etc.) and operators (e.g., all rental housing in the city must be energy efficient at least up to EPC D).

Turning to the relevant **related work on RE**, we note that here RE can help facilitate (weak and strong) stakeholder identification [5, 37, 50] (in complex ecosystems) [47] and working towards agreeing a common goal using such overview building tools as critical systems analysis [34], rich picture sketching [7, 21], or even devising a set of use and misuse scenarios [6].

This may also require a wide set of education and training activities across all levels of the SoS [54, 73].

Requirements for a Cross-Subsystem SLES SoS Governance. While each sub-system will be managed by its respective management structures, there is a need for an additional cross-subsystem governance focused on *inter-dependencies, emergent impacts and rebound effects* that sub-system-specific behaviors could cause in deviating from the agreed-upon goals for and vision of the SLE SoS. Examples of such inter dependencies from the present case study are, for instance, *privacy impacts* from the cross-sub-system aggregation and sharing of householders data, *exacerbated inequalities* (e.g., if the well-off and so well-networked areas of the city are provided with new digital energy services which are not accessible to less affluent areas), *stifling of business opportunities* (e.g., if one sub-system, say the EV charge operators lacks the networking infrastructure to share its data with the others (say energy suppliers) who thus cannot deliver new services), etc.

Presently there is no such governance mechanism in place for the SLE SoS in the Bristol City. However, a similar cross-sub-system impact consideration (focused on identifying and re-dressing the impact of fuel poverty on health) is addressed through setting up a *Health and Well-being Board* [3]. Thus, a process for a cross-subsystem governance body formation should be considered.

The **related work in the RE** discipline has already paid significant attention to (cross-) team coordination and scaling up (software) projects [24, 29, 55], and could provide guidance and advice on this at group, individual and interpersonal levels. RE has also learned to identify and address emerging requirements, e.g., for agile development products [71] and service-oriented systems [9, 72], and similar practices could be applied for the present context.

Requirements for Conflict Resolution must also be considered, as conflicts can emerge both from *incompatible goals* between stakeholders (e.g., the wind turbine installation goal of one community group may conflict with biodiversity preservation goal of another) and *conflicting implementation needs* for technological solutions across sub-systems (e.g., GDPR vs transparent data needs for trusted platforms).

This challenge is centered around the need to align SLES SoS operation with its sub-systems as well as the broader national frameworks. **Related RE research** focused on addressing such negotiation [58, 62], communication [12, 22], policy and regulatory, and standardization [46] needs has long been ongoing. Such techniques as win-win for conflict resolution [44], as well as supporting stakeholders in risk management [14, 69], communication across multidisciplinary and multi-domain teams [38], requirements negotiation [44] and prioritization [42] are well used. All of these are equally relevant for the SLE SoS context.

5.4 Education and Training for a SLE SoS

As discussed in [18], each sub-system within the Bristol City SLE SoS faces a set of its own challenges and each tends to be rather focused on its own internal needs, with little attention given to the other SLE sub-systems or the SoS as a whole. Yet, given that the efficient functioning of the SoS is in the best interests of all SLE stakeholders and citizens, educating these stakeholders on the relevance of the SoS and other sub-systems would also be necessary.

Adopting a Common Vocabulary is a notable challenge in building a common understanding across sub-systems which are converging into a SoS.

As per our Bristol City case study, various sub-system stakeholders use different terms to refer to the same subject and/or the same term to refer to different subjects. For instance, the citizens' representatives discussed 'optimizing consumption' as using as much of their own solar energy as possible, while electricity suppliers 'optimize consumption' by shifting electricity use away from peak demand time. Such terminological misalignment causes misunderstandings and gives the impression of divergent goals between emerging SoS sub-systems and stakeholders.

Related work in the RE has long addressed such terminological mismatches between stakeholders through disambiguation tools [41] developing cross-referenced lexicons, dictionaries [25, 57] and ontologies [10, 27, 43]. This is often done through co-design workshops where colleagues from diverse disciplines/sub-systems collaborate together, or using natural language processing techniques.

General Understanding of Renewable Energy Technologies is another aspect of education and training.

Presently, each sub-system is experiencing a technological flax, as all renewable and clean energy technologies are evolving with an unprecedented speed. As a result, it is difficult for stakeholders as well as the general public to keep up with the innovations or know how to utilize and benefit from them.

Yet, a broad understanding and acceptance of these innovations is essential for a SLE SoS. For instance, a vehicle to grid energy supply service will not be possible without householders and delivery fleet operators buying electric vehicles and agreeing to engage them into this new energy service. Thus, general education and training of all stakeholders and citizens across all of the SLE subsystems on available technologies as well as risk management and financial planning for (domestic and industrial) projects using such technologies is necessary. The requirements analysis would need to identify where and which education and training is needed and how it can be delivered.

6 CONCLUSIONS

As previously noted, the qualitative analysis of the Bristol City ecosystem revealed an ongoing process of convergence of various sectors within the city towards a virtual SLE SoS.

While we also noted a wide variety of RE methods and techniques which can be drawn upon to support the various aspects of this SoS construction activity, this does not at all imply that the RE support for SoS development is fully in place. On the contrary, the nature of RE activities within the SLE SoS is changing: the role of the requirements engineers has broadened from the traditional "elicitation" of the wants and needs of the system owners and users to that of "orchestrating requirements co-creation" within and across the converging sub-systems.

In a recent work, Ncube and colleagues [52] suggested that it is likely that different requirements engineering techniques would be appropriate to support different SoS types. While presently there is no sufficient evidence in RE literature to demonstrate which techniques are relevant to which SoS contexts and types, this case study demonstrates that virtual SoS are in particular need of common vision creation, goal setting and convergence pathways co-design. Yet, these types of SoS have no recognized management or governing body that would support these activities.

In the absence of a clear management body and process, and given the critical importance of energy infrastructure, the responsibility for the SLE SoS is often implicitly assigned to local governments. These, however, are seldom aware of the need for action, as well as poorly qualified to support such a complex technical and technological challenge. Thus, a specialist body for SoS convergence support is much needed for the smart local energy domain. This body would also ideally be supported with a *new type of requirements engineering process for guiding virtual SoS evolution.*

Whichever governance or management body would be defining a common vision and supporting co-design of convergence pathways, it would essentially be shaping the composition of the future SLE SoS, setting the boundaries for it and deciding which prospective sub-systems would be integrated. This places the systems/requirements engineering professionals working on emerging virtual SoS support into positions of excessive power. So questions of power and legitimacy (e.g, whose interests do the engineers further, who empowers the engineers to make decisions, etc.) as advocated for by Critical Systems Heuristics [64, 65] should also be diligently applied and addressed.

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